

Take-Home Project: Sales & Customer Analysis

Project Scenario

You are a **junior data analyst** at an online retail company. The management team wants to understand **sales performance, customer behavior, and regional trends** using historical sales data.

Your task is to **clean, analyze, and extract insights** from the data using **Excel, Python, and SQL**, and present your findings clearly.

Dataset Description

You are given **one CSV file**:
`sales_data.csv`

Columns

Column name	Description
order_id	Unique order identifier
order_date	Date of purchase (YYYY-MM-DD)
customer_id	Unique customer identifier
product	Product name
category	Product category
price	Price per unit
quantity	Number of units purchased
region	Customer region

💡 $\text{Revenue} = \text{price} \times \text{quantity}$

Part 1: Excel Analysis (25%)

Tasks

1. Open the dataset in Excel
2. Perform **data cleaning**:
 - Remove duplicate rows
 - Handle missing values
 - Ensure correct data types
3. Create **Pivot Tables**:
 - Total revenue by **category**

- Total revenue by **region**
- 4. Create charts:
 - Bar chart: revenue by category
 - Line chart: monthly revenue trend

Deliverables

- Cleaned Excel file (`sales_analysis.xlsx`)
 - At least **2 pivot tables** and **2 charts**
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Part 2: Python (Pandas + NumPy) Analysis (35%)

Tasks

Data Loading & Cleaning

- Load the CSV file using Pandas
- Convert `order_date` to datetime
- Create a new column `revenue`

Analysis

Using **Pandas**:

- Total revenue
- Average order value
- Top 5 products by revenue
- Revenue by region
- Monthly sales trend

Using **NumPy**:

- Mean, median, and standard deviation of order revenue
- Identify outliers in order revenue (optional bonus)

Customer Analysis

- Number of unique customers
- Top 10 customers by total spending
- Percentage of revenue from top 20% customers

Deliverables

- Jupyter Notebook (`sales_analysis.ipynb`)
 - Clear comments and markdown explanations
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Part 3: SQL Analysis (30%)

Tasks

1. Create a table called `sales`
2. Insert the data into the database
3. Write SQL queries for:
 - Total revenue
 - Revenue by category
 - Top 5 products by revenue
 - Monthly revenue
 - Revenue by region
 - Customers with total spending above average

Use:

- `GROUP BY`
- `ORDER BY`
- `HAVING`
- `DATE` functions

Deliverables

- SQL file (`sales_queries.sql`) containing:
 - Table creation
 - Insert statements (or import instructions)
 - All queries
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Part 4: Insights & Report (10%)

Tasks

Write a **short report (1–2 pages)** answering:

- Which categories generate the most revenue?
- Which regions perform best?
- Who are the most valuable customers?
- Any noticeable sales trends?
- One business recommendation based on your analysis

Deliverables

- PDF or Word report (`report.pdf` or `report.doc`)
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Final Submission Checklist

Students must submit:

- sales_analysis.xlsx
 - sales_analysis.ipynb
 - sales_queries.sql
 - Short written report
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Grading Rubric

Criteria	Points
Data cleaning	20
Excel analysis	15
Python analysis	35
SQL queries	20
Insights & interpretation	10
Total	100

Academic Integrity

- This is an **individual project**
- Plagiarism or copied notebooks will receive **0 points**

Artificial Intelligence (AI) Usage Policy

1. Purpose

The purpose of this policy is to clarify the acceptable and unacceptable use of Artificial Intelligence (AI) tools (e.g., ChatGPT, Copilot, Gemini) for this project.

AI tools are considered learning aids, not substitutes for understanding, analysis, or independent work.

2. Permitted Use of AI

Students **may use AI tools** for the following purposes:

- Understanding syntax or error messages in Excel, Python, or SQL
 - Clarifying concepts covered in class
 - Getting examples of functions, commands, or queries
 - Improving clarity or grammar of written explanations (without changing meaning)
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3. Prohibited Use of AI

Students **may not**:

- Submit AI-generated code, analysis, or reports **without understanding them**
 - Submit entire solutions generated by AI as their own work
 - Use AI to fabricate data, results, or interpretations
 - Rely on AI during oral defenses or assessments
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4. Responsibility and Accountability

Students are fully responsible for **everything they submit**.

If a student cannot clearly explain or modify their submitted work during an oral defense, the work will be considered **not their own**, regardless of whether AI was used.

5. Oral Defense Requirement

All submitted projects are subject to a **short oral defense (3–5 minutes)**.

During the defense, students must explain:

- Their analysis choices
- Their code and queries
- Their conclusions and recommendations

Failure to demonstrate understanding may result in:

- Partial credit reduction
- Re-submission request
- Grade of zero in severe cases